

DNN-BASED AUDIO SCENE CLASSIFICATION FOR DCASE 2017:DUAL INPUT FEATURES, BALANCING COST, AND STOCHASTIC DATA DUPLICATION



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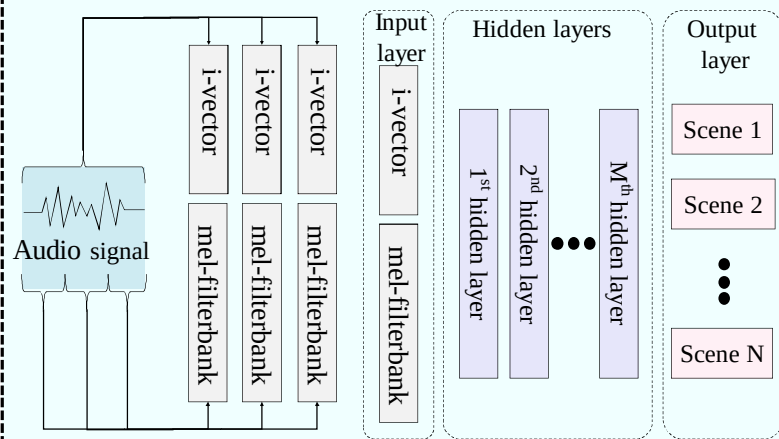
Abstract

- Proposed
 - Dual input features : simultaneously using **two different features** (mel-filterbank energy, i-vector)
 - Balancing cost : **optimized object function** defined for dual input feature approach
 - Stochastic data duplication : DNN **training data manipulation** based on confusion matrices
- Residual architecture was applied with the proposed approaches
- Classification accuracy of **70.6 %** was shown with DCASE 2017 evaluation set

Contribution

- Technique of using two different features were proposed with optimized objective function
- Latest DNN-based advances were applied on audio scene classification

Proposed systems



Dual input features

$$cost = NLL + \alpha \cdot BF_1(W) + \beta \cdot BF_2(W)$$

$\rightarrow$ stops W converging to zeros
 $\rightarrow$ low when the impact of input features nodes are equal

$$f_1(W_x) = \frac{1}{Y} \cdot \sum_{y=1}^Y |W_{x,y}|$$

$$BF_1(W) = Var(f_1(W_1), f_1(W_2), \dots, f_1(W_x))$$

$$f_2(W) = \frac{1}{X} \cdot \frac{1}{Y} \cdot \sum_{x=1}^X \sum_{y=1}^Y W_{x,y}$$

$$BF_2(W) = ReLU(f_2(W^{init}) - f_2(W^{cur}))$$

* $X(x \in X)$: Number of nodes of the input layer
 $Y(y \in Y)$: Number of nodes of the first hidden layer
 NLL : Negative log likelihood
 W : weight matrix between input layer and the first hidden layer
 α, β : hyper-parameters for balancing cost (1000, 100)

Balancing cost

$$E_k = \sum_j C_{j,k} - C_{k,k}$$

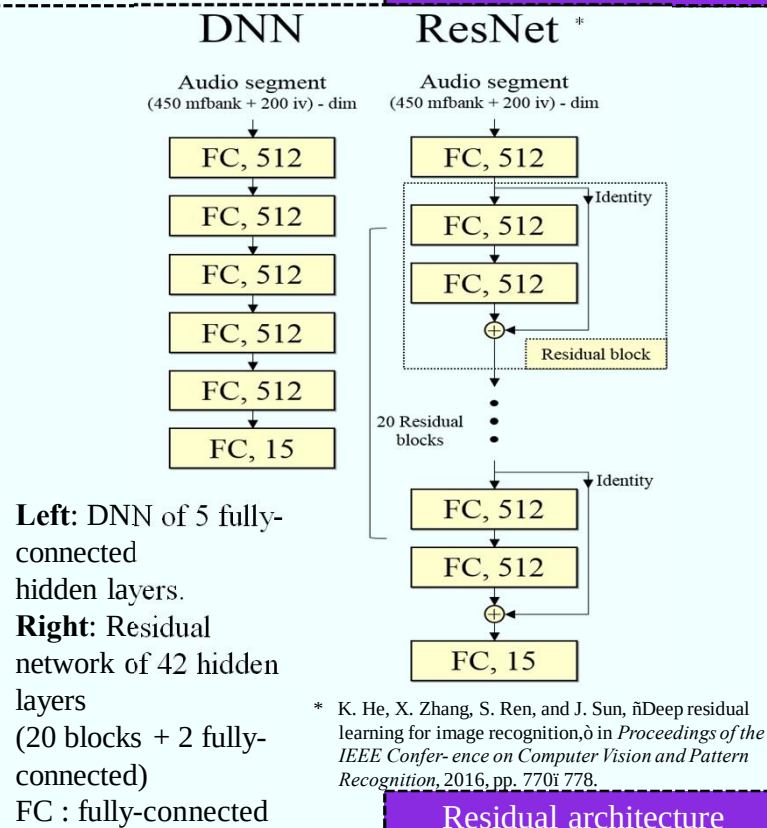
$$A_k = \frac{E_k}{\sum_i E_i}$$

* C : Confusion matrix

E_k : Number of mis-classified segments of class k

A_k : Proportion of data duplication for class k

Stochastic data duplication



Residual architecture

Experiments & Results

- DB : DCASE 2017 task 1
- Dev : 312 segments $\times$ 15 scenes / Eval : 1620 segments
- Feature : 40-dimensional mel-filterbank features + 200-dimensional i-vector
- Dimension of mel-filterbank features were reduced to 10 with LDA, and context frames (left 22, right 22) was applied
- L2-regularization ($\lambda=10^{-4}$), dropout applied

Result (classification accuracy, %)

- System 1: dual input features
- System 2: System 1 + balancing cost
- System 3: System 2 + stochastic data duplication
- System 4: System 3 + residual network

System #	Validation set	Evaluation set
System 1	85.5	67.0
System 2	85.1	66.2
System 3	95.5	67.3
System 4	95.9	70.6